- [x] Learn
- [x] Practice
- [x] Build Logic
- [x] Grow

Python Loops

Repeat tasks, do more with less code

Small Steps Big Programs

Satyarthi

1 What are Loops?

Loops are used to execute a block of code repeatedly. Instead of writing the same code again and again, we can use loops to automate repetition.

2 Types of Loops in Python

1. **for loop** – used to iterate over a sequence (list, tuple, string, range, etc.)
2. **while loop** – used to repeat a block of code as long as a condition is True
3. **Nested loops** – a loop inside another loop
4. **Loop control statements** – break, continue, pass

3 For Loop

Used to iterate over a sequence (like range, list, string).

Code :

```
1  for i in range(5):
2      print(i)
```

Output

```
0
1
2
3
4
```

4 While Loop

Used to repeat a block of code as long as a condition is True.

Code :

```
1  i = 1
2  while i <= 5:
3      print(i)
4      i += 1
```

Output

```
1
2
3
4
5
```

5 Nested Loops

A loop inside another loop.

Code :

```
1  for i in range(3):
2      for j in range(2):
3          print(i, j)
```

Output

```
0 0
0 1
1 0
1 1
2 0
2 1
```

6 Loop Control Statements

Used to change the flow of a loop.

1. **break** – exits the loop completely
2. **continue** – skips the current iteration
3. **pass** – does nothing (placeholder)

7 Examples

Using break

```
1  for i in range(10):
2      if i == 5:
3          break
4      print(i)
```

Output

```
0 1 2 3 4
```

Using continue

```
1  for i in range(5):
2      if i == 2:
3          continue
4      print(i)
```

Output

```
0 1 3 4
```

Using pass

```
1  for i in range(3):
2      if i == 1:
3          pass
4      print(i)
```

Output

```
0 1 2
```

8 Key Takeaways

- [x] Loops help to repeat tasks efficiently.
- [x] Use for loop for sequences.
- [x] Use while loop when condition-based repetition is needed.
- [x] Use break, continue and pass to control the loop flow.
- [x] Nested loops are useful for patterns and complex problems.

Code Learn Improve Repeat

"Repetition today, better solutions tomorrow." – Satyarthi

Keep Practicing Keep Growing

- [x] Learn
- [x] Practice
- [x] Build Logic
- [x] Grow

Python For Loop

Repeat a block of code for a sequence

Simple
Powerful
Versatile

Satyarthi

1 What is a For Loop?

A for loop is used to iterate over a sequence (such as a list, tuple, string, range, etc.) and execute a block of code for each item in the sequence.

It repeats the code a fixed number of times.

2 Syntax

```
for variable in sequence:
    # block of code
```

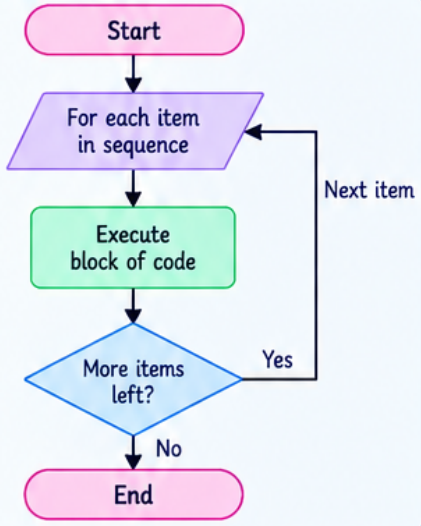
- variable → temporary variable to hold each item
- sequence → list, tuple, string, range, etc.
- code block → runs once for each item

3 How It Works?

1. The loop takes the first item from the sequence and assigns it to the variable.
2. The block of code is executed.
3. The loop moves to the next item.
4. Steps 2–3 repeat until all items are processed.

Stops automatically when all items are completed.

4 Flowchart



5 Example Program

Print numbers from 1 to 5 using for loop.

Code :

```
1  for i in range(1, 6):
2      print(i)
```

Output

```
1
2
3
4
5
```

6 More Examples

1. Iterate over a list

```
fruits = ["apple", "banana",
          "mango"]
for fruit in fruits:
    print(fruit)
```

Output

```
apple
banana
mango
```

2. Sum of numbers

```
total = 0
for i in range(1, 6):
    total += i
print("Sum:", total)
```

Output

```
Sum: 15
```

3. Print characters of a string

```
name = "Python"
for ch in name:
    print(ch)
```

Output

```
P
y
t
h
o
n
```

7 Key Takeaways

- [x] For loop is used to repeat a block of code.
- [x] It works with sequences (list, tuple, string, range, etc.).
- [x] The loop runs a fixed number of times (based on the length of the sequence).
- [x] Useful for processing items one by one.
- [x] Helps in solving real-world problems efficiently.

Did You Know?

You can also use for loop with range(), which is commonly used to repeat code a specific number of times.

"Repeat the right things, build the right results."

– Satyarthi

Keep Learning
Keep Growing

- [x] Solve
- [x] Practice
- [x] Improve
- [x] Become Better

Python For Loop

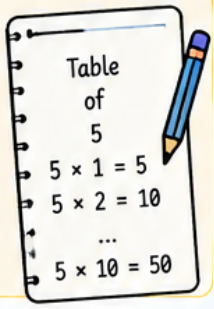
Practice Question

Simple
Powerful
Versatile

Satyarthi

1 Question

Write a Python program to print the tables of a number entered by the user (from 1 to 10) using a for loop.



Sample Input / Output

Input	Output
Enter a number: 5	Table of 5: 5 × 1 = 5 5 × 2 = 10 5 × 3 = 15 5 × 4 = 20 5 × 5 = 25 5 × 6 = 30 5 × 7 = 35 5 × 8 = 40 5 × 9 = 45 5 × 10 = 50

2 Solution Code

Code :

```python
n = int(input("Enter a number: "))
print(f"Table of {n}:")
for i in range(1, 11):
    print(f"{n} x {i} = {n*i}")
```

3 Explanation (Line by Line)

1. Takes the number as input from the user and converts it to an integer.
2. Prints a heading for the table.
3. `for` loop runs from 1 to 10.
4. In each iteration, it prints the multiplication using `n * i`.

4 How It Works?

1. The user enters a number (e.g., 5).
2. The for loop iterates from 1 to 10.
3. In each iteration, the number is multiplied by the current value of i.
4. The result is printed in a formatted way.

5 Key Takeaways

- [x] for loop is useful for repeating a task.
- [x] range(1, 11) generates numbers from 1 to 10 (including 10).
- [x] We can use the same logic for other patterns and calculations.
- [x] Loops help make our code shorter and efficient.

"Practice one problem today, one step closer to mastery." – Satyarthi

- [x] Learn
- [x] Practice
- [x] Build Logic
- [x] Grow

Python For Loop

Practice Question

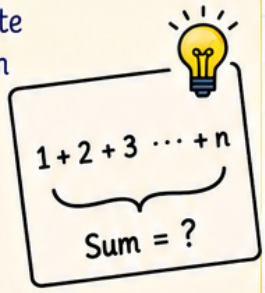
Simple
Powerful
Versatile

Satyarthi

1 Question

Write a Python program to calculate the sum of all numbers from 1 to n (using a for loop).

Take n as input from the user.



Example:

If n = 5, the sum should be:
1 + 2 + 3 + 4 + 5 = 15

2 Sample Input / Output

Input (n)	Output (Sum)
5	15
10	55
1	1
20	210

3 Solution Code

Code :

```
1  n = int(input("Enter a number: "))
2  total = 0
3  for i in range(1, n + 1):
4      total += i
5  print("Sum =", total)
```

4 Example Run

```
Enter a number: 5
Sum = 15
```

```
Enter a number: 10
Sum = 55
```

5 How It Works?

1. Takes input n from the user.
2. Initializes a variable total to 0.
3. The for loop runs from 1 to n.
4. In each iteration, the current number i is added to total.
5. Finally, it prints the sum.

6 Try It Yourself

Modify the above program to:

1. Find the sum of all even numbers from 1 to n.
2. Find the sum of all odd numbers from 1 to n.
3. Calculate the sum of squares ($1^2 + 2^2 + 3^2 + \ldots + n^2$).

Small Steps Big Progress

7 Key Takeaways

- [x] For loop is useful for performing repetitive calculations.
- [x] We can use it with range() to generate a sequence of numbers.
- [x] It helps in solving real-world problems efficiently.
- [x] The same logic can be adapted for different patterns and conditions.

"Practice today, stronger tomorrow."

– Satyarthi

- [x] Learn
- [x] Practice
- [x] Build Logic
- [x] Solve Problems

Python For Loop

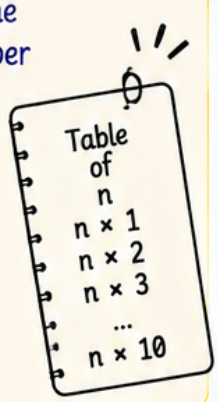
Practice Question

Small Steps Big Programs

Satyarthi

1 Question

Write a Python program to print the multiplication table of a given number using a for loop.



Input:
Take a number n from the user.

Output:
Print the multiplication table of n from 1 to 10.

2 Sample Input / Output

Input	Output
Enter a number: 7	7 × 1 = 7 7 × 2 = 14 7 × 3 = 21 7 × 4 = 28 7 × 5 = 35 7 × 6 = 42 7 × 7 = 49 7 × 8 = 56 7 × 9 = 63 7 × 10 = 70

3 Solution Code

Code :

```
1  n = int(input("Enter a number: "))
2  for i in range(1, 11):
3      print(f"{n} × {i} = {n*i}")
4
```

We use a `for` loop to iterate from 1 to 10.

4 Explanation (Line by Line)

1. Takes input from the user and converts it to an integer.
2. The `for` loop runs from 1 to 10 (including 10).
3. In each iteration, it prints the multiplication result using `n * i`.

5 Try It Yourself

Change the input number and run the program. What do you get for n = 12?

Expected Output (for n = 12)

```
12 × 1 = 12
12 × 2 = 24
12 × 3 = 36
12 × 4 = 48
12 × 5 = 60
12 × 6 = 72
12 × 7 = 84
12 × 8 = 96
12 × 9 = 108
12 × 10 = 120
```

Try with different numbers and observe the pattern!

6 Key Takeaways

- [x] For loop helps us repeat a block of code.
- [x] We can use range(1, 11) to run the loop from 1 to 10.
- [x] It is useful for printing patterns, tables, calculations, etc.
- [x] We can take user input to make the program more flexible.
- [x] This concept is widely used in real-world programming.

"Practice is the key to progress." – Satyarthi

Keep Learning Keep Growing

- [x] Learn
- [x] Practice
- [x] Build Logic
- [x] Solve Problems

Python For Loop

Practice Questions

Small Steps Big Programs

Satyarthi

1 Question 1 – Sum of First n Natural Numbers

Write a Python program to find the sum of first n natural numbers using a for loop. Take n as input from the user. from the user.

Example: If n = 5, the sum should be 1 + 2 + 3 + 4 + 5 = 15.

Solution Code :

```
1  n = int(input("Enter a number: "))
2  total = 0
3  for i in range(1, n + 1):
4      total += i
5  print("Sum =", total)
```

Sample Run

Input	Output
Enter a number: 5	Sum = 15

Explanation

1. Take input n from the user.
2. The `for` loop runs from 1 to n.
3. In each iteration, the value of i is added to `total`.
4. Finally, the sum is printed.

Try Run Modify Explore

2 Question 2 – Print Even Numbers

Write a Python program to print all even numbers from 1 to n (using a for loop). Take n as input from the user.

Example: If n = 10, the output should be: 2 4 6 8 10

Solution Code :

```
1  n = int(input("Enter a number: "))
2  for i in range(2, n + 1, 2):
4      print(i)
```

Sample Run

Input	Output
Enter a number: 10	2 4 6 8 10

Explanation

1. Take input n from the user.
2. The `for` loop starts from 2 and increases by 2 each time (2, 4, 6, ...).
3. It prints all even numbers up to n.

Keep Coding Keep Growing

"Practice today, solve tomorrow." – Satyarthi

Learn
Practice
Build Logic
Solve Problems

Python For Loop

Practice Question

Small Steps Big Programs

Satyarthi

1 Question 3 – Sum of Even Numbers

Write a Python program to find the sum of all even numbers from 1 to n (including n if it is even) using a for loop.

Input:
A number n (e.g., 10)

Output:
The sum of all even numbers from 1 to n.

Example:
If n = 10,
the even numbers are 2, 4, 6, 8, 10
and the sum is 30.

Code
Practice
Repeat
Grow

2 Solution Code

Code :

```
1  n = int(input("Enter a number: "))
2  total = 0
3  for i in range(2, n + 1, 2):
4      total += i
5  print("Sum of even numbers =", total)
```

3 Sample Run

Input	Output
Enter a number: 10	Sum of even numbers = 30
Enter a number: 7	Sum of even numbers = 12

4 How It Works?

1. Takes input n from the user.
2. The `for` loop starts from 2 and increases by 2 each time (2, 4, 6, ...).
3. In each iteration, the value of `i` is added to `total`.
4. Finally, the sum is printed.

Understand Code
Build Logic
Be Better

5 Try It Yourself

1. Run the program for n = 20.
2. What is the output?
3. Modify the program to find the sum of all odd numbers from 1 to n.
4. Can you also print the even numbers while finding the sum?

6 Key Takeaways

- [x] We can use `range(start, stop, step)` in `for` loop.
- [x] This is an efficient way to work with even/odd numbers.
- [x] For loop helps us solve real-world problems easily.
- [x] We can modify the same logic for different conditions.
- [x] Practice more such problems to strengthen your logic.

"Consistent practice creates real progress." – Satyarthi

- [x] Learn
- [x] Practice
- [x] Build Logic
- [x] Improve

Python For Loop

Practice Question

Small Steps Big Programs

1 Question

Write a Python program to find and print the sum of squares of the first n natural numbers using a for loop.
Take n as input from the user.

Example:

If n = 5, the sum of squares should be:
$1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$

Solution Code

```
1  n = int(input("Enter a number: "))
2  total = 0
3  for i in range(1, n + 1):
4      total += i * i
5  print("Sum of squares =", total)
```

Sample Run

Input :

```
Enter a number: 5
```

Output :

```
Sum of squares = 55
```

How It Works?

1. Take input n from the user.
2. The for loop runs from 1 to n.
3. In each iteration, we calculate `i * i` and add it to `total`.
4. Finally, we print the sum of squares.

Key Takeaways

- [x] For loop can be used for mathematical calculations.
- [x] We can accumulate a result using a variable (like `total`).
- [x] This approach is useful for sums, products, and patterns.
- [x] Practice similar problems to strengthen your logic.

"Practice coding, because every problem has a solution."

By Satyarthi

Python
Learn Practice Grow

While Loop

Keep Repeating Until the Condition is False

Small Steps Big Programs

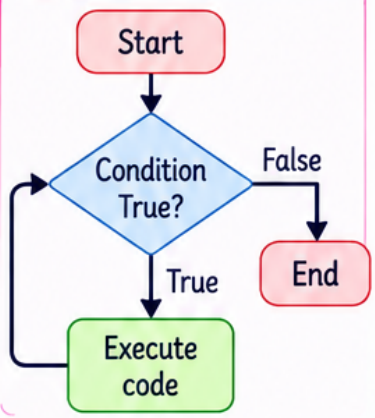
1 What is a While Loop?

- A while loop is used to execute a block of code repeatedly as long as a condition is True.
- The loop stops when the condition becomes False.
- It is useful when we don't know in advance how many times the loop will run.

2 Syntax

```
while condition:
    # code to be executed
    # (repeatedly)
```

3 Flowchart



4 Example

Print numbers from 1 to 5 using a while loop.

```
1  i = 1
2  while i <= 5:
4      print(i)
5      i = i + 1
```

5 Output

```
1
2
3
4
5
```

The loop runs as long as i <= 5 remains True.

6 How It Works?

1. Initialize a variable (e.g., i = 1).
2. Check the condition (e.g., i <= 5).
3. If the condition is True, execute the code inside the loop.
4. Update the variable (e.g., i = i + 1).
5. Repeat the process until the condition becomes False.

7 Key Points

- ☑ The loop runs while the condition is True.
- ☑ Make sure to update the variable inside the loop, otherwise it may run forever (infinite loop).
- ☑ The condition is checked before every iteration.
- ☑ Useful when the number of iterations is not fixed.
- ☑ Can be used for input validation, menus, and repetitive tasks.

8 Real Life Uses

- Taking user input until a valid value is entered.
- Menu-driven programs (e.g., calculator, games).
- Reading data from a file line by line.
- Repeating a task until a certain condition is met (e.g., login until correct password).

"Loops turn small steps into big results."

By Satyarthi

Python

Small Steps Big Programs

While Loop

Practice Question

Practice → Think → Code → Improve

- Learn
- Practice
- Build Logic
- Solve Problems

1 Question

Write a Python program to print all the digits of a given number in reverse order using a while loop.

Take the number as input from the user.

Input / Output Format

Input:

Enter a number: 1234

Output:

Reversed number: 4321

Example

If the input number is 1234, then the output should be 4321.

Try with different numbers like 5678, 100, 908, etc.

Solution Code

```
num = int(input("Enter a number: "))
rev = 0
while num > 0:
    digit = num % 10
    rev = rev * 10 + digit
    num = num // 10

print("Reversed number:", rev)

```

How It Works?

1. Take the number as input from the user.
2. Initialize a variable rev to 0.
3. Use a while loop to extract the last digit using % 10.
4. Add the digit to rev in reverse order (rev = rev * 10 + digit).
5. Remove the last digit from the number using // 10.
6. Repeat until the number becomes 0.
7. Finally, print the reversed number.

"Practice today for a better tomorrow." - Satyarthi

Python

Learn Practice Grow

While Loop

Practice Question

Small Steps Big Programs Bright Future

- Practice
- Improve
- Solve
- Be Better

1 Question

Write a Python program to find the sum of digits of a given number using a while loop.

Take the number as input from the user.

Example

If the input number is 1234,
the sum of its digits is:
1 + 2 + 3 + 4 = 10

Sample Run

Input :

Enter a number: 1234

Output :

Sum of digits = 10

Solution Code

```
1  num = int(input("Enter a number: "))
2  sum_digits = 0
3  while num > 0:
4      digit = num % 10
5      sum_digits = sum_digits + digit
6      num = num // 10
7
8  print("Sum of digits =", sum_digits)
```

How It Works?

1. Take the number as input from the user.
2. Initialize a variable `sum_digits` to 0.
3. Use a while loop to extract the last digit using `num % 10`.
4. Add the digit to `sum_digits`.
5. Remove the last digit from the number using `num // 10`.
6. Repeat until the number becomes 0.
7. Finally, print the sum of digits.

"Practice makes progress." - Satyarthi

Python

Learn
Practice
Grow

While Loop

Practice Question

Small Steps Big Programs

- [x] Think
- [x] Code
- [x] Practice
- [x] Improve

1 Question

Write a Python program to print the multiplication table of a given number using a while loop.
Take the number as input from the user and print the table from 1 to 10.

Solution Code

```
1  n = int(input("Enter a number: "))
2  i = 1
3  while i <= 10:
4      print(n, " x ", i, " = ", n * i)
5      i = i + 1
6
7
8
```

Sample Run

Input :

Enter a number: 5

Output :

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

How It Works?

1. Take the number as input from the user.
2. Initialize a variable `i` to 1.
3. Use a while loop that runs until `i <= 10`.
4. In each iteration, print `n * i`.
5. Increment `i` by 1 (`i = i + 1`).
6. The loop stops when `i` becomes 11.

Key Takeaways

- [x] While loop is useful when the number of iterations is known or unknown.
- [x] Here, we used it to print a multiplication table.
- [x] We can change the range (e.g., 1 to 20).
- [x] Helps in understanding repetition and control flow.

"Practice today, code a better tomorrow."

– Satyarthi